

Day 01 Tasks:

The Scenario:

You are programming a city surveillance drone using an MDP. The reward function gives a constant $+1$ for every second the drone successfully tracks a target vehicle. However, if the drone flies lower than 15 meters over civilian property, it receives a massive single penalty of -100 due to safety regulations.

1. **Mathematical Analysis:** If you configure your agent's discount factor to be **exactly $\gamma = 0$** , how will the drone behave when it detects a target moving down a narrow urban alleyway flanked by low roofs?
2. **Structural Analysis:** If you change γ to **0.99** , how does that alter the drone's long-term operational risk management strategy regarding that same alleyway?
3. **The Markov Trap:** Suppose a tall building temporarily blocks the drone's line-of-sight camera for 3 seconds, losing track of the target's current path. Why does this immediately violate the strict definition of the *Markov Property*, and what structural fix would you apply to the drone's input vector to fix it?

Break down your thoughts on these three points. Don't worry about syntax—focus purely on the mathematical concepts we just established!